

Comparative Study on Different Color Spaces for Segmentation of Acute Leukemia using Automatic Otsu Clustering

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Abstract—Leukemia is common cancer that occurs due to the abnormality in the production of white blood cells in the human blood. This disease which is likely to affect youngsters below the age of 15 years old can be detected by analysing bone marrow samples. However, the process of extracting bone marrow samples is complicated and this could cause patients to experience discomfort, hence blood samples could serve as alternatives since it also contains white blood cell (WBC) information that was needed in determining acute lymphocytic leukemia (ALL). As the development of ALL is quite fast, the early detection of the diseases is vital; hence this segmentation method was developed. The blood images of ALL patients will first be converted from RGB to HSV and Lab color space before the grayscale images produced used as the input to Otsu by clustering technique resulting in black background images while the cells will be staying in white color. Next, the watershed ridge technique has been used to remove the overlapping cells before retrieving the colored version of the WBC images. The results from testing the image segmentation method using a different component from different color spaces showed that the images applied with component b from the Lab color space proved to produce the clearest image of ALL subtypes compared to the other color components applied as the accuracy produced for B cell, T cell and normal cell were at 99.17%, 99.88%, and 99.92% respectively.

Keywords—leukemia, Otsu by clustering, watershed ridge, HSV/LAB color space, ALL subtypes

I. INTRODUCTION

This Leukemia is a type of cancer that is caused by an abnormality in the production of the number of white blood cells that exceed their ideal count in a body. Excess white blood cells will attack other healthy cells. There are several types of leukemia, however, this study will be focusing on acute leukemia. Acute leukemia is more aggressive than chronic leukemia as it will attack suddenly and involve immature white blood cells that have distorted DNA [1].

Leukemia can be detected by analyzing samples from bone marrow, but this method is complicated and invasive.

Analyzing bone marrow samples can give very clear results in diagnosing leukemia, however, the process might lead patients to feel discomfort or infection at the extraction area [2]. Blood sample measurement was used in the analysis to determine the abnormality count of WBC. Blood sample extraction is considered to be less invasive and not complicated to be done. It is easy to acquire blood samples compared to bone marrow samples.

The segmentation of ALL subtypes images and normal cells was required to provide a clear image that was free of undesired cells. It is advisable to not process the entire image at one time as there were areas of the image that are devoid of information, hence only the key segments for image processing by separating the image could be used [3]. There were few image segmentation methods such as region-based segmentation, edge-detection segmentation, and Mask R-CNN that can be used to segment the WBC from the blood samples. However, in this study, Otsu by clustering and watershed ridge algorithm was used to segment WBC from the blood samples images.

Umamaheswari and Geetha [1] presented the segmentation of ALL that starts with resizing and converting the image into grayscale before removing all other components of the blast except for its nucleus. Before the implementation of the nucleus boundary erosion closing technique, undesirable blobs were discarded using the Otsu threshold method. Next, the voids within the nucleus were filled in order to find the largest blob. It was found that from the study, the image segmented works best with a single nucleus compared to many. The suggested technique successfully segments the largest nucleus blob in the obtained blood cell image, assisting hematologists in leukemia detection.

Su, Liu, and Song [4] discussed color segmentation of blood cell images. The Lab color space was chosen because it allows the study of blood cell images in a way that is similar to how the human visual perception system operates. This method uses a k-means cluster to perform initial cell

segmentation, then uses HMRF (Hidden-Markov Random Field) to build a cell image representing model, estimate space representation constraints using Expectation-Maximization (EM) of probability, carry out convergence iteration until optimal, and finally achieve second stage refined segmentation. This process enhances the accuracy and efficiency of the nucleus and cytoplasm segmentation, which was crucial for future cell feature extraction and blast identification for the detection of acute myeloid leukemia.

Mahdy *et. al* [5] convert the input RGB image into the gray level image before applying the Otsu thresholding segmentation technique and implementing a median filter to remove unwanted pixels with the same grey level as infected WBC would be deleted. The image was then complemented to convert infected WBC from black gray level to white gray level to fill the holes in WBC. Otsu in this study used to detect edges of infected WBC to count the level of the blast severity. However, they recommended adding the technique of removing overlapping WBC to increase its accuracy in future studies.

Thus, the current study proposed a segmentation method of ALL subtypes by converting RGB images into grayscale images based on color components involving RGB, HSV, and Lab color space. Next, the Otsu by clustering method was applied in segmenting the resulting image to produce the black background and cells in white color based on clusters determined later. Then, the overlapping cells will be removed using the watershed ridge technique before the result can be analyzed.

II. METHODOLOGY

In this study, blood samples images of ALL patients were used. There were 830 images which include the ALL subtypes which were B cell, T cell, and normal cell used as a dataset in this study. All the images resolution was at 1280×960 pixels and in the format of bitmap. The image segmentation process was run using MATLAB 2020b software.

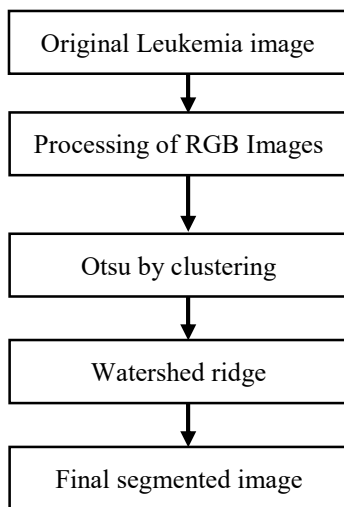


Fig. 1: The block diagram of the proposed method of segmentation for ALL subtypes

A. Preprocessing of RGB Images

Since RGB images, also known as original colors, are impractical in interpreting human-seen colors, segmentation using the HSV color space model and the Lab color space model are employed as parameters in the preprocessing of RGB images [6]. Saturation describes the amount of white light contained in Hue's 0-255 value, with a number closest to "0" indicating greater white light content or saturation. While the brightness value expressed the intensity of light in ranges from "0" or black to "255" white [7].

i) RGB to HSV color space conversion

The equation of converting RGB to HSV was as below.

$$\begin{aligned} V &= Cmax; RD = \text{Red}; \\ GR &= \text{Green}; \text{ and } BL = \text{Blue} \end{aligned} \quad (1)$$

$$H = \begin{cases} 60^\circ \times \left(\frac{GR - BL}{\Delta} \right), & Cmax = RD \\ 60^\circ \times \left(\frac{BL - RD}{\Delta} + 2 \right), & Cmax = GR \\ 60^\circ \times \left(\frac{RD - GR}{\Delta} + 4 \right), & Cmax = BL \\ Undefined, & Cmax = Cmin \end{cases} \quad (2)$$

$$\begin{aligned} Cmax &= \max(RD, GR, BL); \\ Cmin &= \min(RD, GL, BL); \\ \text{and } \Delta &= Cmax - Cmin \end{aligned} \quad (3)$$

ii) RGB to Lab color space conversion

The Lab color space, also known as CIELab, is the most detailed color space model issued by the International Commission on Color Illumination (French: Commission Internationale de l'Eclairage). Where the letters L, a, and b describe the Luminance (Lightness) and Chromatic color values, respectively (A, B) [7].

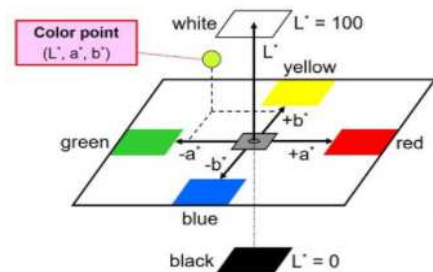


Figure 2: The CIELab color space model [8]

One "channel" for luminance (L) and two channels for color (a and b) make up the CIELab color system. The chromatic a axis in the three-dimensional model runs from green (-a) to red (+a), while the chromatic b axis runs from blue (-b) to yellow (+b). As shown in Figure 2, The lightness

dimension, denoted by L , spans from 0 (complete black) to 100 (complete white) (diffuse white). At the L value of 50, the location where the a and b axes cross is pure, balanced, neutral grey [9].

B. Otsu by clustering

Otsu segmentation is one of the thresholding methods of segmentation. Otsu is used to construct the graph-based image threshold or reduce the intensity in the binary image automatically. The image has two types of pixels, resulting in two media which were background and foreground. The appropriate isolation threshold for these two categories can then be determined [2, 5, 10, 11]. The grey level is made up of several shades of grey ranging from black at the lowest intensity to white at the highest intensity, with values ranging from 0 to 255 [12].

Using Otsu by clustering, the background and the foreground can be differentiated as the cluster was set according to the color space chosen. For instance, if there were 4 clusters, one cluster will be set at 255 while the remaining clusters will be set at 0. The purpose for this setting is because in the color space as the numbers set to 0, it will produce black color while 255 will produce the white color. Hence, 255 is known as the setting for white cell images and 0 is the setting for black background colors. The specific number of clusters for each color space tested for ALL_B cell image was as shown in Table 1. Using this method, the binary images of white '1' and black '0' which represent the WBC and the background were produced [12].

C. Watershed ridge

Table 1: The clusters of ALL B cell images run for Otsu by clustering method

Colour Space Cluster	R	G	B	h	s	v	l	a	b
1	255	255	0	0	0	255	255	0	255
2	0	0	0	0	0	0	0	0	0
3	0	0	255	255	255	0	0	0	0
4	-	0	0	0	-	-	0	255	-
5	-	-	0	-	-	-	0	-	-

Watershed ridge is used as the transform method for the last step of this image segmentation method. It can be applied to a picture to transform it from one domain to another, allowing for rapid feature extraction and recognition. The area with less grey variation has a lower gradient; however, the grey changes at the adjacent boundary for different purposes are greater than the dissimilarity of pixels within the region [2, 13]. As a result, during the processing of the watershed transform, the pixel gradient has a bigger influence than the pixel grey. In most cases, the referred image for segmentation is the gradient image.

Based on Fig. 3, a catchment basin is formed by pixels that assemble in the course of a same minimum. The raised

areas that divide the several catchment basins are known as watersheds. The catchment basins are the partitions required, and the watersheds are the boundaries between them [2]. From Fig. 3, the watershed lines can be pictured as the boundary line for each WBC meanwhile the catchment basin can be seen as WBC in the blood samples tested [14].

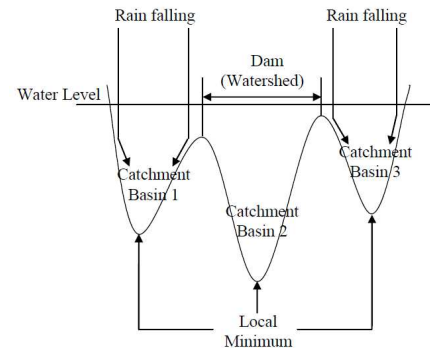


Fig. 3: Illustration of watershed algorithm [12]

Watershed ridge was used as a topography to produce the final segmentation images that was improved by lessening the noise along with the background [15]. The Otsu clustering image will be used as the source for the watershed ridge approach. The watershed ridge method will then remove any overlapping cells that are near to one another, resulting in a final segmented image of WBC from the blood sample.

D. System Performance

Since the image segmentation method developed will automatically segmented the blood samples in determining ALL subtypes, the system needed to be evaluated. The accuracy, sensitivity and specificity of the system will be evaluated.

Accuracy is the degree to which a measure is near to an acceptable value. Sensitivity or recall measures the part of positive case which actually measured as positive in a system. Specificity is the opposite to sensitivity in which it measured the part of negative cases which actually measured as negative [16].

The formula for accuracy, sensitivity and specificity are as shown below.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

TP : True positive
 TN : True negative
 FP : False positive
 FN : False negative

$$Sensitivity = \frac{TP}{TP + FN} \quad (6)$$

$$Specificity = \frac{TN}{TN + FP} \quad (7)$$

True positive is defined as when the model predicts the positive class properly. Meanwhile, true negative is known as a result in which the model correctly predicts the negative class. A false positive occurs when the model forecasts the positive class inaccurately. A false negative is an outcome in which the model forecasts the negative class inaccurately [16].

III. RESULTS AND DISCUSSION

In the presentation of results, a comparison between original RGB images, HSV color model, and Lab color model were provided as the preprocessed images to Otsu by clustering. At that point watershed ridge algorithm has been conducted in order to recognize which color component will

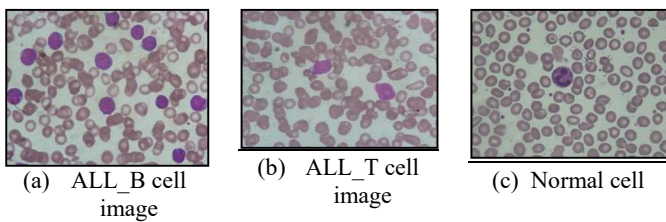


Fig. 4: Original images of ALL subtypes and normal cell

produce the best image segmentation for acute lymphoblastic leukemia. All the images of ALL subtypes and normal cell used are in normal condition as shown in Figure 4.

Hue, saturation, and value components (from HSV color space), red, green, and blue components (from RGB color model), and luminance, a, and b components (from Lab color model) were used as the input into Otsu by clustering. All of these color components were compared in order to

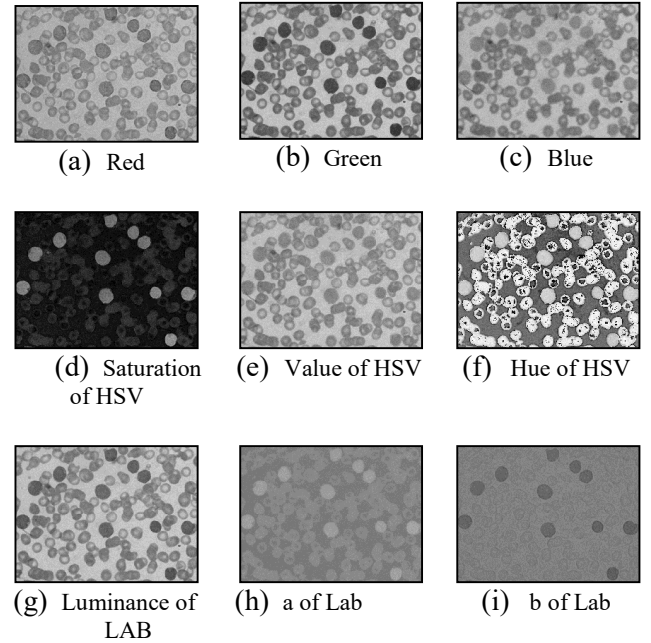


Fig. 5: Resultant images of ALL_B cell after being converted from RGB to HSV/LAB color component

determine the optimal color components for image segmentation of ALL subtypes.

As seen in Figure 5 the result from blue, green, value, luminance, and b are as shown in the image (b), (c), (e), (f), (g), and (i) respectively have their ALL subtypes emerges in darker color compared to the red blood cells and the whiter background. However, as for red, hue, and a color component, the ALL subtypes seem whiter in color compared to the red blood cells and their surroundings.

To get the black background and white image for WBC, the Otsu by clustering segmentation was performed to the images after they were processed from RGB to HSV and Lab. The image resultant as shown in Figure 6 from red, green, blue, value, hue and luminance images in (a), (b), (c), (e), (f) and (g) appear to have a not very clear image of WBC since RBC can still be seen in the image. Saturation, a and, b component images in (d), (h), and (i) shows the WBC appears to be clearer than other color component. Few parts of the nucleus of WBC were missing as it had similar pixel as the RBC however the image can still be used for the input of watershed ridge technique. The images developed by different color components have their own exact number of clusters based on the individual images in order to achieve an image with a black background and the interested cells in white color.

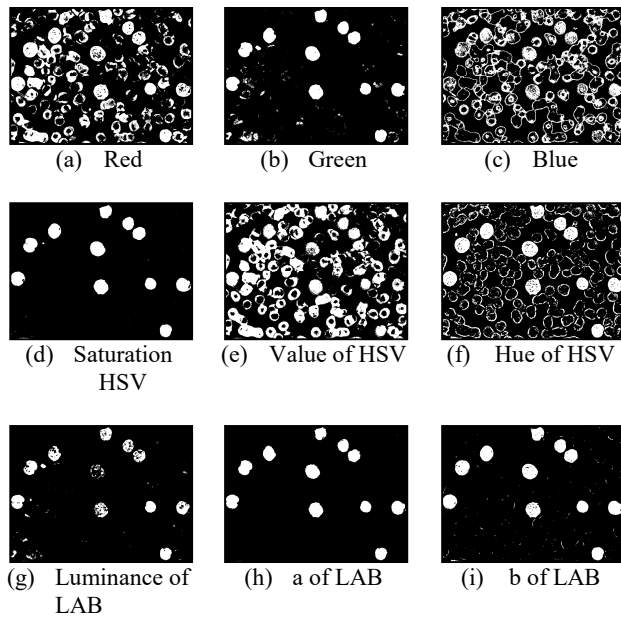


Fig. 6: Resultants images of applying Otsu by clustering on ALL_B cell to different color components

Watershed ridge were then applied after Otsu by clustering process were done. This process is done to remove overlapping cells that still appear in the images produced by Otsu clustering.

The Figure 7 (ALL_B cells), Figure 8 (ALL_T cells) and Figure 9 (normal cells) showed the retrieved image after watershed ridge method was applied to each color components.

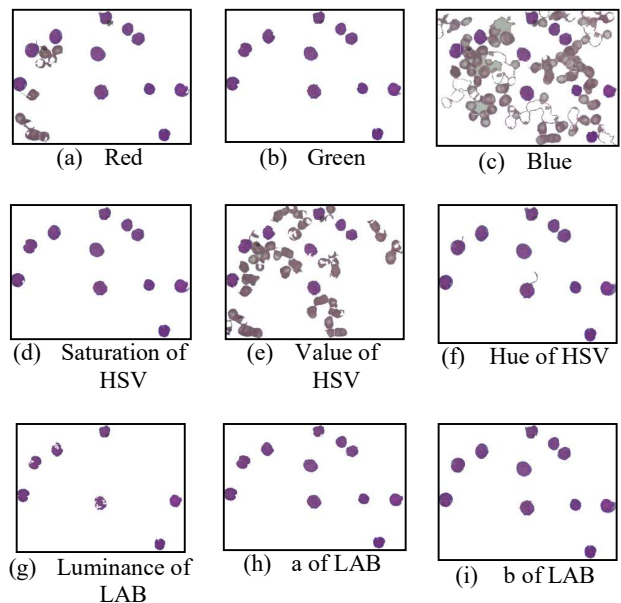


Fig. 7: Results of images for ALL_B cell after watershed ridge technique applied to different color components

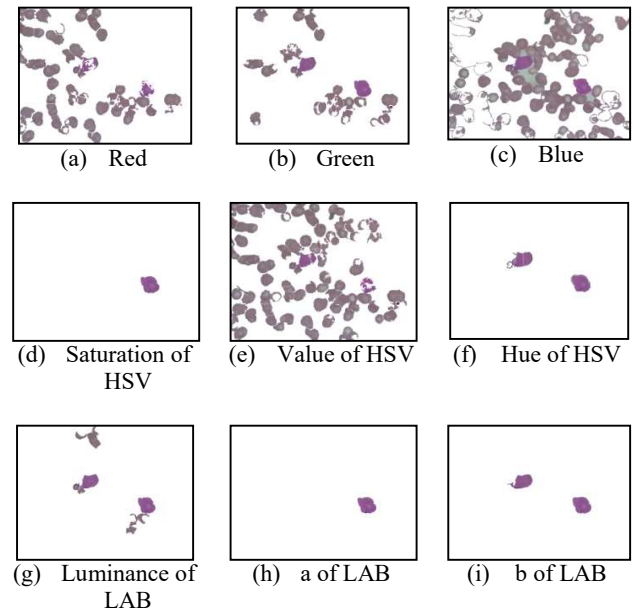


Fig. 8: Resultant images for ALL_T cell after watershed ridge technique applied to different color components

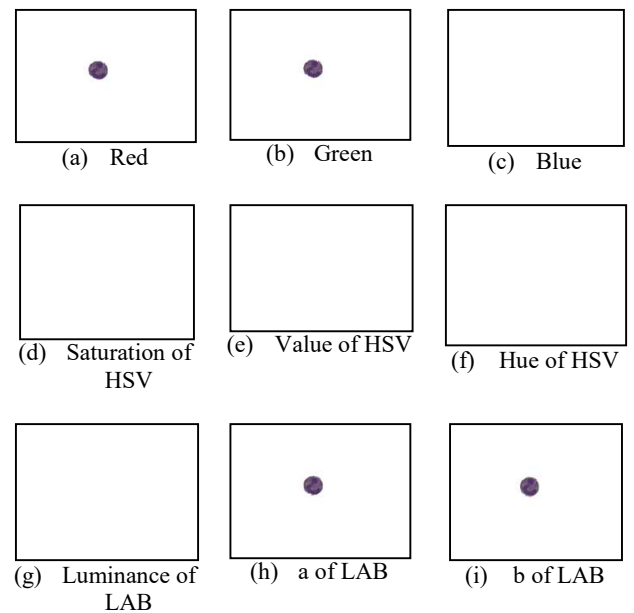


Fig. 9: Resultant images for the normal cell after watershed ridge technique applied to different color components

The method did not work out well for red, green, blue, value, hue, luminance and A component in images (a), (b), (c), (e), (f), (g) and (h) as there were still RBC, uncleaned background removal and the detail of the WBC images produced is not clear. These are due to the RBCs image that has similar pixel as ALL subtype cells.

Table 2: The performance of image segmentation method using B component from LAB color space

	Accuracy	Sensitivity	Specificity
B cell	99.17%	89.53%	99.97%
T cell	99.88%	97.22%	99.93%
Normal cell	99.92%	95.49%	99.97%

Table 2 shown the results of the image segmentation performance of ALL subtypes (Figure 4) that used b component of Lab color space, where it produced the highest percentage of accuracy of the three images of a B cell, T cell and a normal cell at 99.17%, 99.88% and 99.92% respectively. The sensitivity of the image segmentation evaluated were at 89.53%, 97.22% and 95.49%, while the specificity obtained were 99.97%, 99.93% and 99.97% respectively.

IV. CONCLUSION AND RECOMMENDATION

In conclusion, the segmentation of ALL subtypes and normal cell were done in order to find the best acute leukemia images with the implementation of various color component. Based on the analysis, the image segmentation produced from the color component b of Lab color space was proven as the clearest and best image compared to another color component as the accuracy produced for ALL subtypes B cell, T cell and normal cell (from Figure 4) were at 99.17%, 99.88% and 99.92% respectively. The image segmented with b component of LAB color space contains a very detailed characteristics of the white blood cells which were the ALL subtypes. Hence, these images can be useful information for future study which could be feature extraction of the WBC images in order to develop a classifier that can help the medical officer in diagnosing ALL.

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REFERENCES

- [1] D. S. G. D. Umamaheswari, "Segmentation and Classification of Acute Lymphoblastic Leukemia Cells Tooled with Digital Image Processing and ML Techniques," in *Proceedings of the Second International Conference on Intelligent Computing and Control Systems (ICICCS 2018)*, 2018: IEEE.
- [2] D. Virmani, N. Jain, K. Parikh, and S. Upadhyaya, "Boundary Outlier Centroid Based Reduced Overlapping Image Segmentation," *Journal of Engineering Science and Technology Review*, vol. 11, no. 5, pp. 1-9, 2018, doi: 10.25103/jestr.115.01.
- [3] P. Sharma. "Computer Vision Tutorial: A Step-by-Step Introduction to Image Segmentation Techniques (Part 1)." Analytics Vidhya. <https://www.analyticsvidhya.com/blog/2019/04/introduction-image-segmentation-techniques-python/> (accessed February 20, 2022).
- [4] J. Su, S. Liu, and J. Song, "A segmentation method based on HMRF for the aided diagnosis of acute myeloid leukemia," *Comput Methods Programs Biomed*, vol. 152, pp. 115-123, Dec 2017, doi: 10.1016/j.cmpb.2017.09.011.
- [5] K. A. E. Lamia Nabil Mahdy, Ashraf Darwish, Aboul Ella Hassannien, "Automatic Counting of Infected White Blood Cells Using Multi-Level Thresholding," presented at the IEEE Ninth International Conference on Intelligent Computing and Information Systems (ICICIS), 2019.
- [6] L. X. Liu Feng, Chen Yi, "An efficient detection method for rare colored capsule based on RGB and HSV color space " presented at the 2014 IEEE International Conference on Granular Computing (GrC), 2014.
- [7] S. A. R. Thomas Agung Setyawan, Suhendro, Helmy, Arif Nursyahid, Ari Sriyanto Nugroho, "Comparison of HSV and LAB Color Spaces for Hydroponic Monitoring System," in *Proc. of 2018 5th Int. Conf. on Information Tech. Computer. and Electrical Engineering (ICITACEE)*, 2018: IEEE.
- [8] J. Beetsma. "The CIELAB L*a*b* System – the Method to Quantify Colors of Coatings." Prospector. <https://knowledge.ulprospector.com/10780/pc-the-cielab-lab-system-the-method-to-quantify-colors-of-coatings/> (accessed).
- [9] "Defining and Communicating Color: The CIELAB System." [Online]. Available: <https://cdn-s3.sappi.com/s3fs-public/sappi/Defining%20and%20Communicating%20Color.pdf>
- [10] C. Narsimha Raj Kasu, "Segmentation on Chest Radiographs using Otsu's and K-means Clustering Method," in *Proceedings of the International Conference on Inventive in Computing Applications*, 2018: IEEE.
- [11] K. A. P. Divya, "Segmentation of Defected Regions in Leaves using K-Means and Otsu's Method," presented at the 4th International Conference on Electrical Energy Systems (ICEES), 2018.
- [12] H. A. H. Sharifah Lailee Syed Abdullah, Nursuriati Jamil, "Segmentation of Natural Images Using an Improved Thresholding-based Technique," presented at the International Symposium on Robotics and Intelligent Seasons 201, 2012.
- [13] L. S. Chen Jia Xin, "A Medical Image Segmentation Method Based on Watershed Transform," presented at the Proceedings of the 2005 The Fifth International Conference on Computer and Information Technology (CIT'05), 2005.
- [14] E. U. Francis, "Design of A Screening System For Acute Leukemia Cells Based on Bone Marrow Samples," School of Mechatronic Engineering, Universiti Malaysia Perlis, 2012.
- [15] L. M. S. Romel Bhattacharjee, "Detection of Acute Lymphoblastic Leukemia using Watershed Transformation Technique," presented at the 2015 International Conference on Signal Processing, Computing and Control (2015 ISPC), 2015.
- [16] S. Mishra. "Baffling Concept of True Positive and True Negative." Towards Data Science. <https://towardsdatascience.com/baffling-concept-of-true-positive-and-true-negative-bfb340f107> (accessed March 20, 2022).